2024 Calculus II Supplement Homework (3)

1. Which of the following series converge absolutely, which converge conditionally, and which diverge ? Give your reasons.

(a) $\sum\limits_{n=1}^{\infty} \frac{(-1)^n n}{2n^2+n+1}$. (b) $\sum\limits_{n=2}^{\infty} \frac{(-1)^n}{n^p (\ln n)^q}$ $(p > 0,\ q > 0)$. (c) $\sum\limits_{n=1}^{\infty} \frac{(-1)^n}{n^{p+\frac{1}{n}}}$ $(p > 0)$.

2. Exercise 50, (1),(2),(3),(7) in page 29 from Calculus Supplement Homework.

3 Let $\sum\limits_{n=0}^{\infty} \frac{a^n}{n!} x^n$ and $\sum\limits_{n=0}^{\infty} \frac{b^n}{n!} x^n$ be power series. Find $\left(\sum\limits_{n=0}^{\infty} \frac{a^n}{n!} x^n\right)\left(\sum\limits_{n=0}^{\infty} \frac{b^n}{n!} x^n\right)$ and find all x such that the equality holds.

4. Suppose that a power series $\sum\limits_{n=0}^{\infty} c_n x^n$ with its the radius of convergence $R > 0$ and its sum function $f(x)$ is a odd (even, resp.) function. Prove that $c_n = 0$ for every even (odd, resp.) number n.